Corollary 6.3 (Quartic Onset of Alignment Degradation)**.** *Fix a skill S_i and let $\theta(t)$ be a fine-tuning trajectory satisfying the assumptions of Theorems 6.1 and 6.2. Then there exists $t_0 > 0$ such that for all $t \in [0, t_0]$, the utility loss for skill S_i satisfies $\Delta u_i = \Omega(\lambda \gamma^2 t^4)$.*

The quartic law provides a quantitative explanation for rapid safety collapse observed in practice [25, 44]. Thus, our results show that this perceived sudden failure is actually a smooth quartic due to curvature coupling.

We lastly note that the preceding results are stated with respect to the sensitive subspace M_i defined by the Fisher information at θ^*. In practice, $F_i(\theta)$ varies along the fine-tuning trajectory, inducing rotation of the corresponding eigenspaces. Under mild local regularity, this rotation can be controlled via standard Davis-Kahan perturbation bounds, and the proofs extend with only constant-factor and lower-order corrections to the drift and loss bounds. We provide the formal statement and proof in Appendix B.

7 Experiments

Our theory establishes that alignment degradation is geometrically inevitable under the AIC, with the Fisher information matrix (FIM) characterizing the curvature structure that governs this fragility. We now validate (1) that the FIM exhibits low-rank structure required by the AIC, and (2) that our notion of geometric overlap successfully predicts which fine-tuning tasks will degrade alignment.

7.1 Low-Rank Fisher Structure

Our theory operates in weight space, where the FIM's eigenvector define alignment-sensitive directions (Definition 3.4). However, prior empirical work characterizes alignment through low-rank structure in representation space (activations), rather than weight space [3, 60, 65, 66]. Specifically, these works show that for a linear layer $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, the average gradient $\nabla L_W(T)$ with respect to alignment sequences T typically concentrates in an r-dimensional subspace of the representation space, with $r \ll \min\{d_{\text{out}}, d_{\text{in}}\}$.

This distinction is crucial because representation-space low-rank identifies which neurons are sensitive, while weight-space low-rank via FIM quantifies how much curvature exists along sensitive directions (λ in our theory). These are complementary perspectives on the same underlying geometry, but our theory relies on the FIM formulation to characterize second-order drift (Theorem 6.2). In Appendix A.4, we prove that these two notions are related: if $\nabla L_W(T)$ concentrates in an r-dimensional representation space, then the FIM inherits low-rank, establishing AIC Property 1. Thus, existing empirical evidence implies our weight-space Fisher structure, but we validate this directly.

Computational Approach. Computing the full FIM for billion-parameter models is intractable. Following [66], we use block-wise analysis, treating each module separately. For a weight matrix $W \in \mathbb{R}^{d \times d}$, we first flatten to $\mathbb{R}^{d^2}$, then project to a low-dimensional space using Rademacher random projection [25, 30, 41, 61]. We then compute gradients on 100 safe completions from BeaverTails [28] and form the projected FIM.

Results. In Figure 2, we visualize sorted eigenvalues for W_{up}, W_{gate} and W_{down} across all MLP blocks (with the remaining modules deferred to Appendix D.1). Notably, the eigenvalue decay exhibits the sharp low-rank structure leveraged by our theory. This validates AIC Property 1 empirically and confirms that the connection from representation-space to weight-space low-rank holds in practice.

7.2 Geometric Overlap and Misalignment

Recent work documents emergent misalignment (EM): fine-tuning on seemingly benign datasets causes safety degradation [6, 57]. [59] explain this through "persona shifts" identified via Sparse Autoencoders (SAEs) [8, 15, 21] in representation space, a diagnostic approach that requires access to a target evaluation set to identify which personas are problematic. We demonstrate that our geometric framework offers a complementary and predictive explanation. Specifically, EM arises from geometric overlap between fine-tuning updates and the Fisher-defined alignment subspace, quantifiable *before* observing safety failures.

We approximate the utility loss from Proposition 3.3 using an **Overlap Score (OS)**. For each fine-tuned model, we extract per-module weight changes $\Delta W \in \mathbb{R}^{d \times d}$, flatten to $\mathbb{R}^{d^2}$, apply the same Rademacher projection $P_{\text{random}} \in$

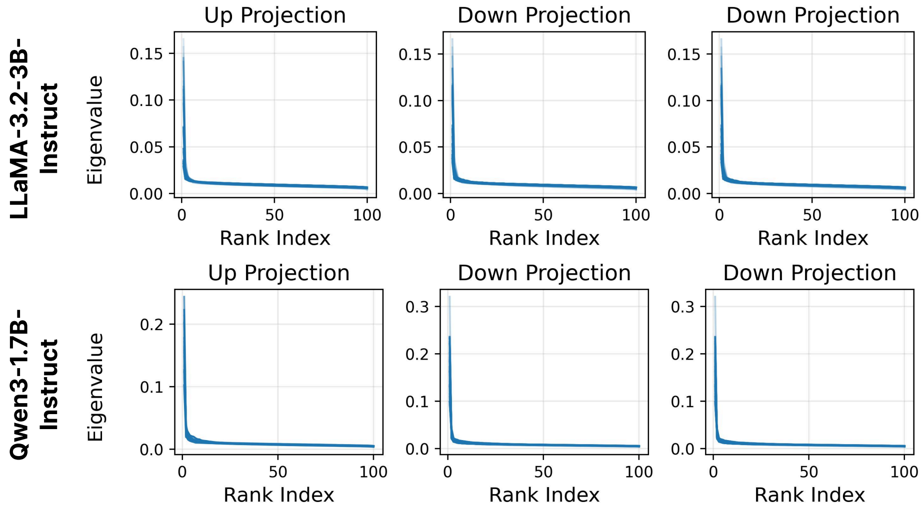

Figure 2: Top eigenvalues of FIM approximated over 100 random samples from *BeaverTail*'s safe subset. Each subplot consists of multiple lines in different transparency levels for different layer indices, with each layer showing a similar low-rank structure.

$\mathbb{R}^{4096 \times d^2}$ used for the FIM, and compute:

$$\mathbf{OS} = \left(P_{\text{random}} \text{vec}(\Delta W)\right)^\top \mathbf{F}\left(P_{\text{random}} \text{vec}(\Delta W)\right).$$

This measures how much the fine-tuning update projects onto the high-curvature alignment subspace. Higher OS predicts greater alignment degradation, enabling prospective risk assessment.

Experimental Setup. We fine-tune Qwen3-1.7B-Instruct [53] and LLaMA-3.2-3B-Instruct [23] using both LoRA [27] and full fine-tuning on datasets in three categories: (1) *Benign* (SamSum, Alpaca Random 100, GSM8K Random 100), (2) *Seemingly Benign* (Alpaca Top 100 via gradient matching [25], Risky Financial Advice, Bad Medical Advice), and (3) *Harmful* (Pure Bad [44]). Following standard protocols [25, 58, 65], we evaluate using Gemini-2.5-Flash [13] as a judge to assign Harmfulness Scores (HS, 1-5 scale) on 520 AdvBench queries [67]. Full dataset and training details in Appendix D.

Table 1 showcases that the three dataset categories produce distinct harmfulness levels, with *Seemingly Benign* datasets achieving HS scores comparable to explicitly harmful data. Critically, all fine-tuning increases harmfulness relative to the base model, but the magnitude is strongly modulated by dataset characteristics. This confirms our framework's prediction that alignment degradation is unavoidable but scales with task-geometry coupling. In full fine-tuning, our Overlap Score successfully distinguishes dangerous from safe tasks (Figure 3). The *Seemingly Benign* datasets exhibit OS patterns similar to *Pure Bad*, while truly benign datasets show substantially lower overlap. This validates that geometric coupling plays a role in degradation: datasets with no apparent overlap with harmful content still break safety when they couple strongly to alignment-sensitive parameters. One exception is *Alpaca Top 100*, where OS does not clearly separate from benign datasets, likely due to (1) noise in random projection to lower dimensions, or (2) Alpaca's diversity compared to domain-specific datasets.

Limitations. We additionally note that LoRA does not show the same clear OS-to-HS correlation (Figure 3, right panels). We hypothesize this occurs because LoRA introduces "intruder dimensions" [50] that shift the model's top eigenvectors, making overlap with the pre-training Fisher less reliable. This supports our theoretical result on parametrization dependence. Additionally, the first-order OS may be insufficient for LoRA because its constrained update structure amplifies the induced coupling condition (AIC Condition 3), requiring second-order curvature analysis via $\nabla g(\theta^*)$. Computing this remains computationally prohibitive [33]; developing tractable curvature estimation is critical future work.

In spite of this limitation, we note that while [59]'s SAE-based approach identifies what changes in representations, our geometric framework explains *why* these changes occur (curvature forces drift into $\mathcal{M}_i$) and predicts which tasks will

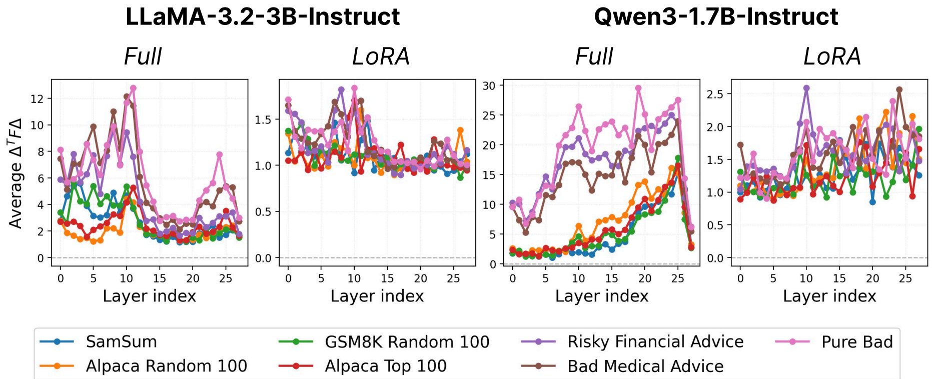

Figure 3: Average Overlap Score per Transformer Block of 7 Fine-Tuning Datasets. Each value represents the average OS of all modules in the block.

cause them via OS. Critically, our approach requires only the fine-tuning dataset and pre-computed FIM—no access to target evaluation sets or trained SAEs—making it applicable for prospective risk assessment before deployment.

Table 1: Harmfulness Score (HS) across fine-tuning datasets

Finetuning Dataset	**Qwen3-1.7B-Instruct**		**LLaMA-3.2-3B-Instruct**	
	LoRA	**Full**	**LoRA**	**Full**
No Fine-tuning	1.24		1.11	
Benign				
SamSum	2.07	2.15	1.10	1.77
Alpaca Random 100	2.68	2.92	1.31	1.76
GSM8K Random 100	3.63	3.42	1.10	1.38
Seemingly Benign				
Alpaca Top 100	3.15	3.68	1.18	3.70
Risky Financial Advice	3.94	3.78	4.71	3.82
Bad Medical Advice	3.92	3.86	2.46	3.65
Harmful				
Pure Bad	4.55	3.98	4.89	4.51

8 Discussion & Implications

We have established that alignment degradation is an intrinsic geometric property of gradient-based fine-tuning, not an artifact of adversarial attacks or poor data curation. The AIC reveals why first-order defenses fail: while projecting updates into null spaces provides temporary protection, second-order curvature effects systematically bend trajectories back into sensitive subspaces. Our quartic scaling law explains the rapid onset of safety failures observed empirically and demonstrates that current defenses are fundamentally insufficient because they ignore the dynamic, curved geometry of the parameter space. Moreover, our framework enables practical diagnostic tools for safer fine-tuning. We propose that practitioners can estimate the coupling parameter γ before training to assess risk, monitor the projection size during training for early warning signals, and audit these values after training to quantify alignment degradation. More fundamentally, effective preservation of alignment requires curvature-aware methods that track evolving sensitive subspaces, constrain second-order acceleration into these regions, and balance competing curvature constraints across multiple alignment skills.

Our results challenge the assumption that alignment preservation is primarily an engineering problem solvable